$$Q = (W^Q)^T X \qquad K = (W^K)^T X \qquad V = (W^V)^T X$$

$$\text{Attn}(Q, K, V) = V \cdot \text{softmax}\left(\frac{K^T Q}{\sqrt{d_K}}\right)$$

1a) The attention mechanism takes the output position being generated and computes relevance between that and every input position. Tokens that are more relevant get higher weights. The model learns through training which dims attention to those tokens

b) The query encodes the question and projects a dedicated query space. It specifically optimizes for the task of searching for relevant context

c) Key is the source of the query, the dot product Q·K measures alignment. Separate value projections allow for each role to be computed

d) The value is the information of a token and contents. A separate value projection provides more flexibility and extracts different information.

e) Different heads in Transformers use attention to understand context. The position in the pipeline are different resulting in different representations.

2a)

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{matrix} 1 & 0 \\ 1 & 1 \end{matrix}$$

b) $\sqrt{d_k} = \sqrt{2}$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

c)

$$\begin{bmatrix} 0.5 & 0.3703 \\ 0.5 & 0.6297 \end{bmatrix}$$

d) $Z = VA$

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0.5 & 0.3303 \\ 0.5 & 0.6697 \end{bmatrix} = \begin{bmatrix} 1 & 0.6606 \\ 1.5 & 2.0091 \end{bmatrix}$$

e) q1 attracts both to q2, but the other 2 keys

3a) $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 0.7311 \\ 0.2689 \\ 0.0900 \end{bmatrix}$ $0.7311 + 0.2689 + 0.0900 = 1$

$$\frac{1}{e^2+e^1+e^0} \begin{pmatrix} e^2 \\ e^1 \\ e^0 \end{pmatrix}$$

b) $\begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}$ $\begin{bmatrix} 0.8725 \\ 0.118 \\ 0.016 \end{bmatrix}$ more sharper at the top

c) There is very little information from the bottom two words making learning harder.

4a) $X = 512 \times 6$

$W^a = 512 \times 64$

$a = 64 \times 6 = W^{a^T} X$

b) $K^T Q = 6 \times 6$

c) 6×6

d) $(64 \times 6)(6 \times 6) = 64 \times 6$

5a) $256 \times 64 = 16384$

$3 \times 16384 = 49152$

$49152 \times 4 = 196608$

b) $256 \times (4 \times 64) = 65536$

$= 262144$

c) $6 \times 262144 = 1572864$

d) 4X

6a) Single head attack only copies a single query and only probes one relationship. With multi-head, each head is able to ask different queries and obtain different relationships. A single head sacrifices Q diversity with multi-fits. Parameters stay same so no additional cost as well.

b) i) It could mean that the model is not using the multi; part of the multi-head attention properly but could be fine if it really is actually important to them.

ii) It forces the model to not rely on any single head. When a head is randomly removed, the remaining heads have to pick up the slack.